

Product Popularity Prediction using Online Retail Dataset

Group – C7

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This project leverages transactional data from the Online Retail dataset to analyze and predict product popularity using Machine Learning techniques. In the competitive landscape of e-commerce and retail, understanding customer preferences and predicting demand patterns has become essential. With an increasing focus on personalized marketing and inventory optimization, identifying popular products early can help retailers enhance customer satisfaction, reduce overstock or stockouts, and ultimately maximize profits.

The dataset comprises detailed records of invoices, product descriptions, quantities sold, unit prices, invoice dates, and customer identifiers, primarily from a UK-based retail store. We will begin by performing extensive data cleaning removing duplicates, handling missing values, and filtering out negative or erroneous transactions. The cleaned data is then aggregated at the product level to compute a variety of descriptive statistics, such as total quantity sold, total revenue generated per product, average unit price, and the number of unique invoices involving each product.

To classify products based on their popularity, we employ a threshold-based labeling mechanism where products are considered 'popular' if their total quantity sold is above the median across all products. Using these computed features and binary popularity labels, we build and train supervised classification models, including Random Forest, Decision Tree, and Logistic Regression, to predict whether a given product is likely to be popular.

The models are evaluated using key performance metrics like accuracy, precision, recall, and F1-score to ensure reliability and generalizability. Additionally, we perform feature importance analysis to determine which attributes most influence product popularity, offering valuable insights into consumer behavior.

The final model can serve as a critical tool for demand forecasting, allowing businesses to make informed, data-driven decisions regarding product assortment, promotional strategies, and inventory management. To make the solution more accessible and user-friendly, we present the results through a lightweight web interface built using Streamlit (or Flask). This interface allows users to interact with model predictions, explore product-level insights, and visualize sales trends through dynamic charts and tables. The web app bridges the gap between technical analysis and business decision-making, making the insights actionable for non-technical stakeholders.

